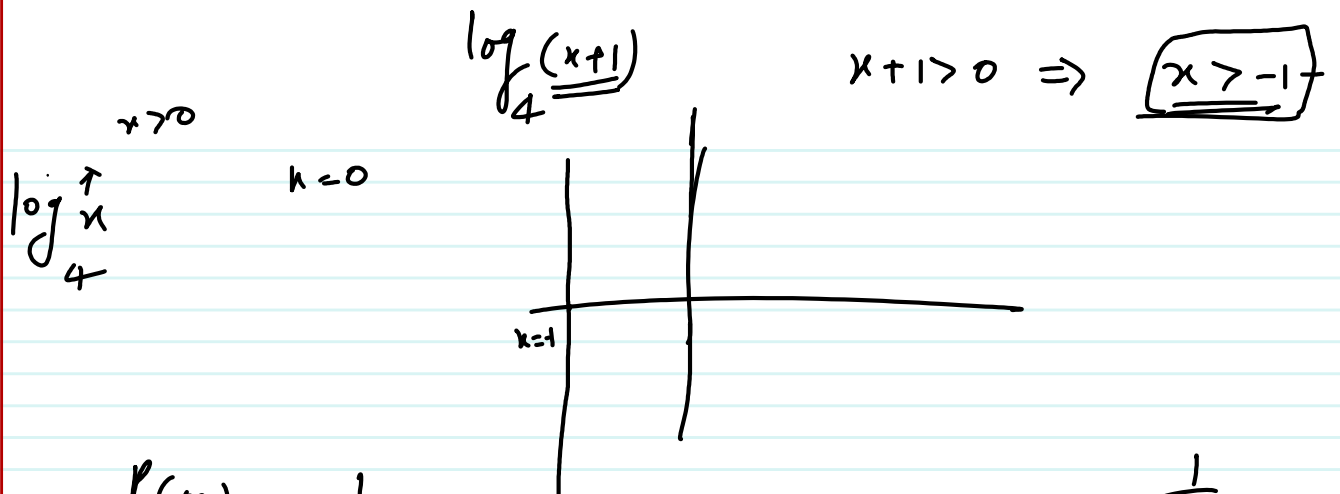




$$f(m) = \log_{10} m+1 - \frac{1}{2} \log_{m+1} (0.01) \quad \frac{1}{100}$$

$$\textcircled{> 10}$$

$$= \log_{10} m+1 - \frac{1}{2} \log_{m+1} (10)^{-2}$$

$$\underline{m > n > 9}$$

$$\textcircled{m > n}$$

$$= \log_{10} m+1 + \log_{m+1} 10 \quad \textcircled{\frac{1}{\log_{10} m+1}}$$

$$g(n) = \underline{x + \frac{1}{n}} \rightarrow n_1 > n_2$$

$$g(n_1) > g(n_2)$$

$$g'(n) = 1 - \frac{1}{n^2}$$

$$f(m) > f(n)$$

$$(\Leftrightarrow)$$

$$\boxed{g'(n) > 0}$$

$$\underline{g'(n) < 0}$$

$$\boxed{a^{\log_b c} = \log_b a^c}$$

$$m \log_3 2 + 2 \log_3 m = 16$$

$$\boxed{2 \log_3 m + 2 \log_3 m = 16}$$

a

$$a^{\log_b c} = y$$

$$\log_b c \log_a a = \log_a y$$

$$\frac{1}{\log_c b} = \frac{1}{\log_y a} \log_b c = \log_a c \Rightarrow \boxed{\log_{10} 1} = 0$$

$$y = a^{\log_b c}$$

taking

$$\log_{10} y = \log_b c \log_{10} a$$

$$= \frac{\log_{10} c}{\log_{10} b} \times \log_{10} a$$

$$= \left(\frac{\log_{10} a}{\log_{10} b} \right) \times \log_{10} c = \log_b a \cdot \log_{10} c$$

Taking both 10 to b

$$\log_{10} y = \log_b a \log_{10} c$$

$$\Rightarrow y = c^{\log_b a}$$

$$a^{b^c} = \underline{\underline{(a^b)^c}}$$

$$\log_{10} y = \log_b a \times \log_{10} c$$

$$\underline{\underline{y = c^{\log_b a}}}$$

$$\log_{10} (1 + (n+1)(n+2)(n+3)(n+4)(n+5)(n+6))$$

$$n=3 \quad \log_{10} 1 = 0^8$$

$$f(x) = x^2 + 2$$

$$g(x) = \frac{x}{2}$$

$$\underline{(f+g)(x)} = f(x) + g(x)$$

$$\left(\frac{f}{g}\right)(x) = 3 \times \frac{f(x)}{g(x)}$$

$$(3f)(x) = (3 \cdot f(x))$$

$$0 \rightarrow x^2 \rightarrow \text{line} \rightarrow f(x) = \underline{x+1} \quad \underline{3x+2}$$

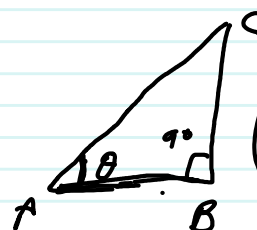
trigo

$$x \times \frac{360}{\pi} =$$

$$\underline{\underline{\sin \theta}} \quad \underline{\underline{\cos \theta}}$$

$$\sin \theta = \frac{BC}{AC}$$

$$\cos \theta = \frac{AB}{AC}$$



$$\theta \times \frac{2\pi}{360}$$

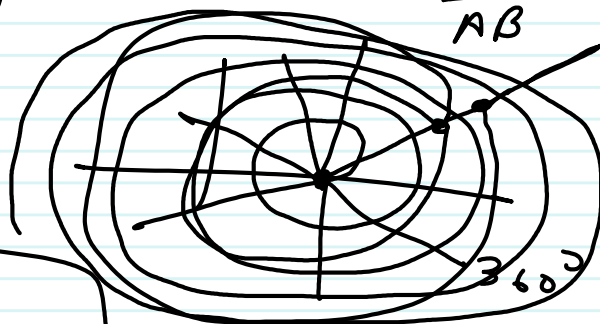
$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{BC/AC}{AB/AC} = \frac{BC}{AB}$$

$$\frac{3}{2} \times \frac{360}{\pi} = 540^\circ \quad \left(\frac{3}{2}\pi\right)$$

$$\theta = 60^\circ$$

$$360 + 360$$

$$\leq 720$$



$$\sin 0^\circ = 0$$

$$\cos 0^\circ = 1$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}}$$

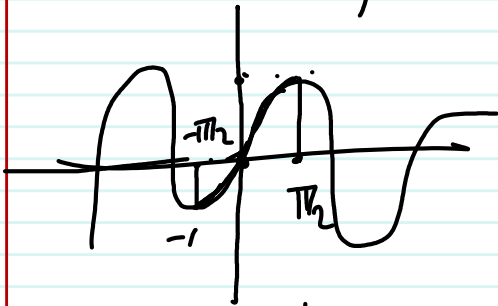
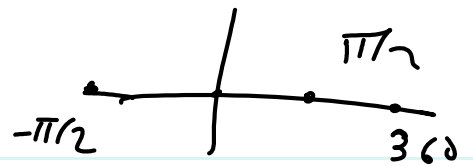
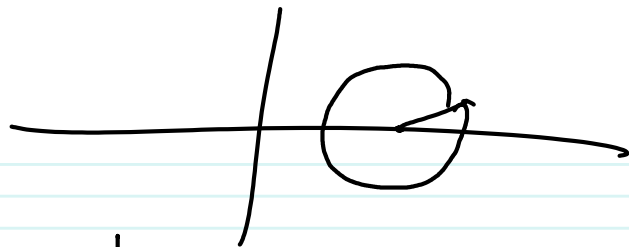
$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\sin 90^\circ = 1$$

$$\cos 90^\circ = 0$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$



$$\underline{\underline{(-\pi/2, \pi/2)}}$$

$\sin(x)$

$$\sin \theta =$$

sec cot

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cot = \frac{1}{\tan \theta}$$

$$f(x) = \sin(x + \pi/2)$$

$$\theta \times \frac{\pi}{360^\circ} \rightarrow$$

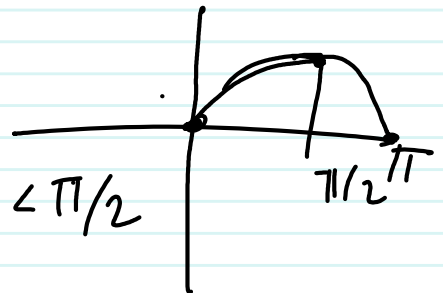
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$30^\circ \times \frac{\pi}{360^\circ} = \left(\frac{\pi}{12}\right)$$

$$f(x) = \log(\sin x)$$

$$0 < x < \pi/2$$



$$\pi = 3.14159...$$

cos



$$\sin^2 x + \cos^2 x = 1$$

$$\sin 2x = 2 \sin x \cos x$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 1 - 2\sin^2 x \end{aligned}$$

$$\sin(\theta + \alpha) = \sin \theta \cdot \cos \alpha + \cos \theta \cdot \sin \alpha$$

Sequence

$(-1, 1, 3, 5, 7, \dots)$

$$f: \mathbb{N} \rightarrow \mathbb{R}$$

$$f(n) = 2n - 1$$

$n \in \mathbb{N}_0$

$$f(1) = 1$$

$$f(3) = 5$$

$$f(2) = 3$$

$$f(4) = 7$$

$(1, 3, 5, 7, 9, 11, 13, 15, 17, \dots)$
 $n=0 \quad n=1 \quad n=2 \quad n=3$

$$f(n) = ()$$

$(1, 3, -1, 0, 5, - \dots)$

$$f(n) = 3n + n^2$$

$(0, 4, 10, \dots)$

$n=0$

$(0, 4, 10, \dots)$

$n=0 \quad n=1 \quad n=2 \quad n=3 \quad \dots$

$$f(n) = 3n + n^2$$

3

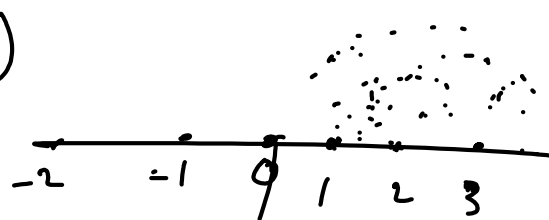
$$f: \mathbb{N} \rightarrow \mathbb{R}$$

$f(0) = 0$

$()$

$\uparrow$

$$f(x) = \frac{x-3}{x+1} \quad (\mathbb{R} \setminus \{-1\})$$



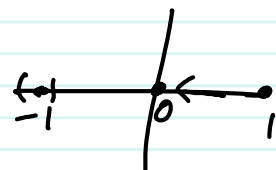
$$f(n) = \frac{n-3}{n+1}$$

$$\mathbb{N} \subset \mathbb{R} \setminus \{-1\}$$



$$f(1) = \frac{1-3}{1+1} = \frac{-2}{2} = -1$$

$$f(n) = \left(\frac{1}{n+1} \right)$$



$$(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{1000})$$

$$f(n) = \frac{n-3}{n+1} \quad n=10 \quad \lim_{n \rightarrow \infty} f(n) = 1$$

$$\frac{n-3}{n+1} = \frac{n}{n+1} - \frac{3}{n+1}$$

$$= \frac{n}{n(1+\frac{1}{n})} - \frac{3}{n+1}$$

$$= \frac{1}{1+\frac{1}{n}} - 3 \times \frac{1}{n+1}$$

$$\frac{1}{\infty} = 0 \quad \frac{1}{\infty} = 0 \quad \frac{1}{\frac{1}{0}} = 0$$

$$1 - 0 = 1 \quad 3 \times 0 = 0$$

$$1 + \frac{1}{h} \rightarrow 1 \quad \frac{1}{h} \rightarrow 0$$

$$n \rightarrow \infty, \frac{1}{h+1} \rightarrow 0$$

$$\frac{1}{h} \rightarrow 0$$

$$\frac{1}{h+2} \rightarrow 0$$

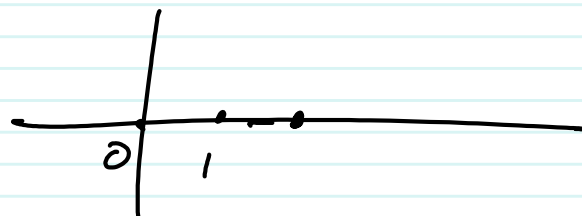
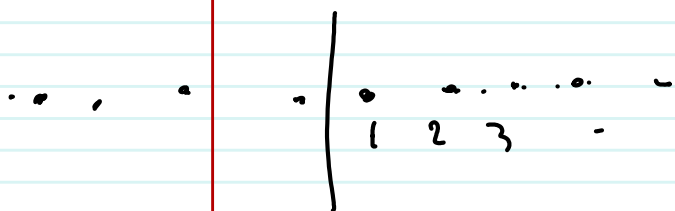
$$n \rightarrow \infty \frac{1}{h+1} = \left(\frac{1}{3}\right) \quad \mathbb{N} \rightarrow$$

$$1 + \frac{1}{h} \rightarrow 1$$

$$\frac{1}{1 + \frac{1}{h}} \rightarrow 1$$

$$f(x) = \frac{x-3}{x+1}$$

$$\rightarrow \frac{2-3}{2+1} = -\frac{1}{3}$$



$$f(x) = \frac{x-1}{x^2-1}$$

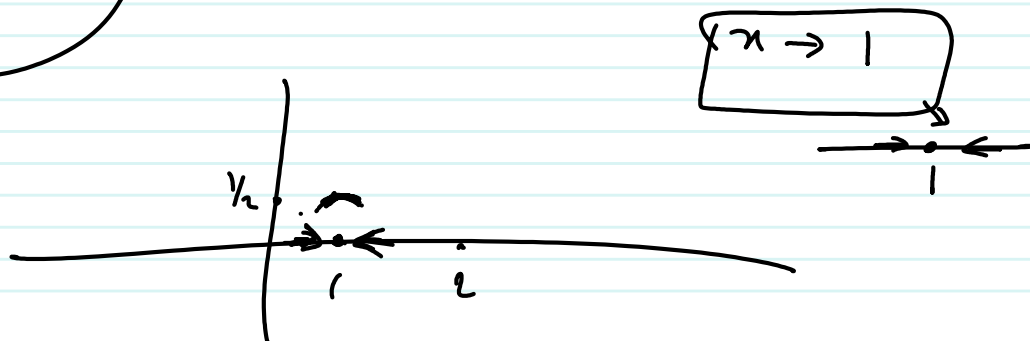
$$x=1 \quad x \rightarrow 1$$

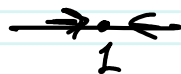
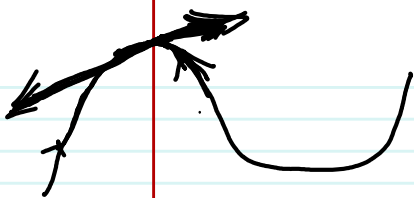
$$\lim_{n \rightarrow \infty} \frac{\sin n}{n} \rightarrow 0$$

$$\frac{\cos n}{1} = 1$$

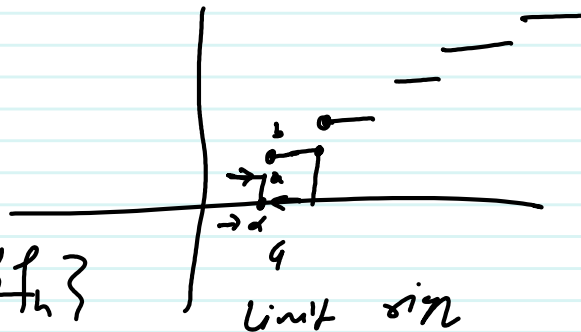
$$= \frac{(x-1)}{(x+1)(x-1)}$$

$$= \frac{1}{x+1} = \frac{1}{2}$$





$$f(n) \rightarrow \{f_n\}$$



$$\{a_n\} \xrightarrow{n \rightarrow \infty} \underline{a}$$

$$\boxed{f(n) = x^2}$$

$$a_n = \frac{1}{n} \rightarrow 0$$

$$f(a_n) = (a_n)^2 \rightarrow \frac{a^2}{n^2}$$

$$\textcircled{f_n} = a_n^2 \Rightarrow 0^2 = 0$$

$$a_n \rightarrow a$$

$$\boxed{\log a_n \rightarrow \log a}$$